Universität Rostock

Traditio et Innovatio

Introduction to Demographic Methods

(1.2: Introduction – additions)

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Additional topics, figures, and links for topic 1 (introduction)

Compositional effects

- In demography, it refers to changes in overall population rates (like death, birth, or productivity) driven by shifts in the structure of the population —such as age, gender, or education— rather than changes in the behavior of specific subgroups.
- These effects, also known as demographic structure impacts or population makeup changes, explain how an aging population increases crude death rates even if medical care improves.
- Usage Examples in Demography
  - Demographic Dividend: An increase in the share of the working-age population (a compositional change) can boost economic output per capita.
  - Mortality Rates: A country with a higher proportion of elderly people may have a higher crude death rate, even if it has better healthcare than a younger country.

Person years contributed from Preston et al., 2000

Each horizontal line represents one life line for a specific individual.

Each vertical line represents a quarter in the year.

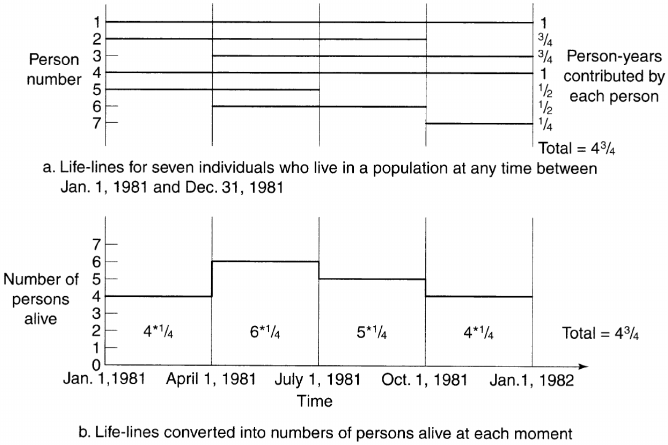


Figure 1.1 Demonstration of the equivalence of the two methods for recording person-years

See more in: Preston et al., 2000, P 5

Importance of using denominators to calculate rates

It is important to use proper "denominators" to calculate rates. This helps put raw counts in a correct perspective.

See one example in this video from national news where the 6 digit precise count of babies born in 2025 is given which is reduced by 3.4% in comparison to 2024.

The first question is, what is the denominator here? Is it based on the number of women in reproductive ages? Did we have a change, increase/decrease, in their population?

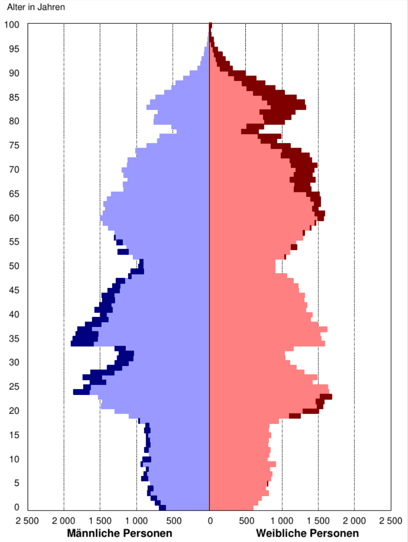
See more in: https://www.youtube.com/watch?v=2I-W7M78Eog&t=724s

Population pyramid of Mecklenburg-Vorpommern (MV) in 2024

Each horizontal line represents one age group and the count of men (left, blue) or women (right, red) in that group.

See more on page 41 of: https://rathaus.rostock.de/de/rathaus/aktuelles_medien/statistisches_jahrbuch_2025_erschienen/371689

Abb. 2.05 Altersstruktur der Bevölkerung der Hanse- und Universitätsstadt Rostock am 31. Dezember 2024

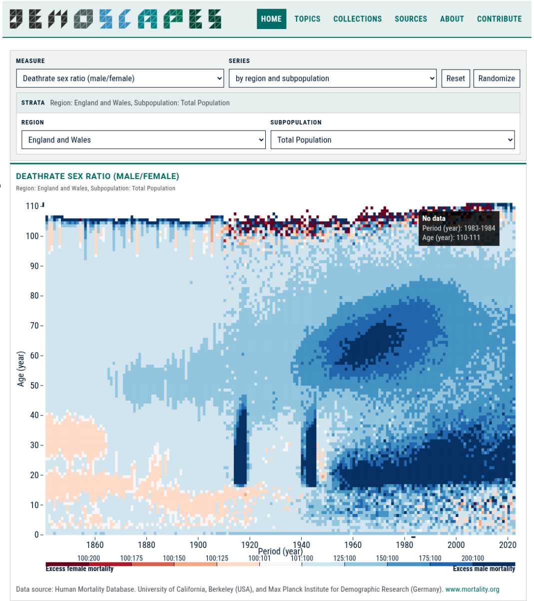


Hanse- und Universitätsstadt Rostock, Hauptamt, Kommunale Statistikstelle

Lexis diagram, and Lexis surface

Visit this website, prepared by Jonas Schöley (MPIDR) and colleagues, for a collection of Lexis diagram and Lexis surfaces (what was their difference according to Wachter, 2014 book's chapter 2?!):

One example on the right, see more here: https://demoscapes.org



Switch to Ernesto Amaral's slides: Lecture01a

Hands-on part

Needed installations (or checking requirements)

A) For topic 1, I requested you to go over Git, version control, and GitHub course. Did you face any errors or have questions? Do you know how to clone, fork, and create a GitHub repository? If yes, pull the latest changes from the course's repository.

B) For topic 2, I shared instructions on how to install R, RStudio, and use Renv for handling required packages. If you are done with those, go to step C below.

C) For today's hands-on part, see the Readme file under "03_MonicaALabs" folder and download the requested files under "data" folder.

We will go over the notebook called "1_intro_edits.qmd" and see what it does. You need to activate Renv and install R packages listed in the first code block of the notebook, i.e., "tidyverse, here, readxl, janitor" (please check the notebook for the latest list).

Needed: R, RStudio, and Renv environment.

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the "key reading material".
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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References

- Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.
- Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.